

A thermally formed flavin radical pair that forms fast enough to matter cannot sense the geomagnetic field

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Abstract

Whether an Earth-strength magnetic field can bias an enzymatic reaction through the radical-pair mechanism depends on two properties of the radical pair: the exchange coupling J between its electrons and its lifetime τ . Both are usually treated as free parameters. For a flavin radical pair that returns to a closed-shell ground state by back electron transfer, they are not independent: a single electronic coupling t fixes the back-transfer rate ($\propto t^2$) and, through superexchange with the ground state, the exchange coupling ($|J| = 2t^2/\Delta$), so that $(|J|/\hbar)\tau$ depends only on the driving force Δ and the reorganization energy λ . We calibrate t , λ and the hyperfine couplings for the flavin–amine pair proposed for monoamine oxidase by density functional theory, and anchor the lifetimes to measured flavin radical-pair recombination in photolyase. Detailed balance then closes the question for any enzyme that operates in the dark: a pair that forms thermally as fast as the enzyme turns over must lie within $\Delta \leq 0.93$ eV of the ground state and recombines in $\leq 3.9 \times 10^{-11}$ s, with $|J| \geq 2.9 \times 10^5$ MHz. Even with the exchange coupling set to zero and the lifetime constraint ignored, the dipolar coupling at the contact distance, summed over the computed spin distributions, limits the geomagnetic response to $\leq 4.3 \times 10^{-4}$ %. Cryptochrome escapes both constraints because the photon supplies Δ : its separated radical pair retains a 0.18–1.42 % orientation-averaged response in the same model.

Introduction

The radical-pair mechanism (RPM) is the only experimentally supported way in which a magnetic field as weak as the geomagnetic field ($\sim 50 \mu\text{T}$) can alter the yield of a chemical reaction.^{1,2} It underlies the leading model of avian magnetoreception, in which photoexcitation of the flavin in cryptochrome creates a spin-correlated flavin–tryptophan radical pair.^{3–5} Whether the same mechanism could operate in the flavoenzymes that metabolize neurotransmitters—monoamine oxidases A and B (MAO-A/B) and D-amino-acid oxidase

(DAO)—has been raised repeatedly,^{6,7} and would couple an environmental field directly to monoamine and amino-acid neurotransmitter levels.

General bounds on weak-field effects are known. The magnitude of a primary RPM effect in biochemistry is limited by electron spin relaxation to 10^{-3} – 10^{-4} or less,⁸ and interrational exchange and dipolar couplings suppress singlet–triplet interconversion unless the radicals are separated by more than ~ 3.5 nm, or by ~ 2 nm where the two interactions partially cancel.⁹ What these bounds leave open is whether the chemistry of a particular enzyme allows its radical pair to reach the parameter regime they require. That question turns on the exchange coupling J and the lifetime τ , and in the enzyme literature both are usually chosen independently: a small J is paired with a long τ , or the reverse.

They cannot be chosen independently. It is established that the exchange coupling of a radical ion pair measures the electronic coupling of its charge recombination,¹⁰ and the exponential distance dependence of J is commonly taken to follow that of the squared electron-transfer coupling.⁹ Here we make the link quantitative for the case that matters for a flavoenzyme: a radical pair whose recombination partner is the closed-shell ground state. One coupling t then sets both the exchange coupling and the lifetime, and their product depends only on the driving force and the reorganization energy. Combined with detailed balance for a pair that must form thermally at the rate the enzyme turns over, this confines the radical pair of any dark-operating flavoenzyme to lifetimes below 3.9×10^{-11} s and exchange couplings above 2.9×10^5 MHz. We calibrate the inputs for the flavin–amine pair proposed for monoamine oxidase, show that the contact-distance dipolar coupling limits the weak-field response by itself, and contrast the result with cryptochrome, where light removes the constraint.

Model and conventions

Spin Hamiltonian

The radical pair is described by the Hamiltonian (frequency units)

$$H = \omega_B (\hat{n}_B \cdot \mathbf{S}_1 + \hat{n}_B \cdot \mathbf{S}_2) + \sum_k a_k \mathbf{S}_{e(k)} \cdot \mathbf{I}_k + J \mathbf{S}_1 \cdot \mathbf{S}_2 + 2D (\mathbf{S}_1 \cdot \hat{n})(\mathbf{S}_2 \cdot \hat{n}) - \frac{2D}{3} \mathbf{S}_1 \cdot \mathbf{S}_2, \quad (1)$$

with $\omega_B = g\mu_B B/h$, isotropic hyperfine couplings a_k of nucleus k on radical $e(k)$, and $\hat{n}$ the interrational axis. Two conventions are stated explicitly because they differ between literature sources. First, with the exchange term written as $J \mathbf{S}_1 \cdot \mathbf{S}_2$, $J = E_T - E_S$ is the full singlet–triplet splitting; in the convention of Efimova and Hore,⁹ $\langle S|H_J|S \rangle = J_{\text{EH}}$ and $\langle T|H_J|T \rangle = -J_{\text{EH}}$, so that $J = -2J_{\text{EH}}$. Second, the dipolar term is the full point-dipole tensor, and with the field along $\hat{n}$ the $T_{\pm}$ and T_0 levels are split by D ; this is the D of ref 9, $D = -77.9 \text{ MHz}/(r/\text{nm})^3$. Both statements are verified numerically: with $B = 0$ and no hyperfine terms the triplet levels lie at $D/3$ (twice) and $-2D/3$ relative to their centroid, and the singlet lies J below it.

Master equation

The spin density operator of a pair born in the singlet evolves as

$$\dot{\rho} = -2\pi i[H, \rho] - \frac{k}{2}\{P_S + P_T, \rho\} + \sum_{i=1,2} \frac{2}{T_2^e} \left(S_{iz} \rho S_{iz} - \frac{1}{4} \rho \right), \quad (2)$$

with symmetric recombination $k = 1/\tau$ and single-electron transverse relaxation time T_2^e . The time-integrated singlet yield $\Phi_S = k \int_0^\infty \text{Tr}[P_S \rho] dt$ is obtained by exact solution in Liouville space, and the magnetic field effect is $\text{MFE} = [\Phi_S(B) - \Phi_S(0)]/\Phi_S(0)$ at $B = 50 \mu\text{T}$. Nuclei are treated as spin- $\frac{1}{2}$ except where a spin-1 ^{14}N model is stated.

Hyperfine couplings

For the flavin anion radical we use the isotropic couplings of N5 (523 μT) and N10 (189 μT).¹¹ Our density functional calculation of the Fermi-contact couplings (B3LYP/def2-TZVP on GFN2-xTB geometries) gives 441 and 165 μT , within 16 % and 13 % of these values, and we use the same method for the partner radicals, for which no measured couplings are available. The largest coupling in the benzylaminium radical cation, a model for the MAO-B substrate, is 257 MHz, from an $\alpha\text{-CH}_2$ proton hyperconjugated with the nitrogen-centred singly occupied orbital; the value is conformation dependent. For the alaninium model of the DAO substrate it is 58 MHz. Because the conclusions below are set by couplings between the electrons, the partner coupling changes the size of the residual response but not the conclusion: over partner couplings of 5–360 MHz the contact-distance bound of section lies between 3.6×10^{-8} and 8.8×10^{-5} % (Supporting Information).

Calibration

Electronic coupling at the active-site contact

The putative MAO radical pair is the flavin anion radical and the aminium radical cation of the substrate. Back electron transfer moves an electron from the flavin singly occupied orbital (the LUMO of oxidized flavin) to the amine singly occupied orbital (the HOMO of the neutral amine, its nitrogen lone pair). We compute the coupling $t = \langle \text{LUMO}_{\text{Fl}} | F | \text{HOMO}_{\text{am}} \rangle$ by fragment-orbital projection onto the Kohn–Sham Fock matrix of the dimer (B3LYP/def2-SVP),¹² with a methylamine placed on the flavin si face with its nitrogen on the ring normal through N5 and its lone pair directed at N5. Across 33 geometries (N5–N distance 3.0–6.0 Å, four orientations, steric clashes excluded) the coupling decays exponentially with $\beta = 2.5\text{--}3.4 \text{ Å}^{-1}$ for t^2 , the value expected for through-space tunnelling, and at the 3.5 Å contact distance $t = 109\text{--}290 \text{ meV}$ (geometric mean 160 meV). This is a controlled model of the

substrate position, not a docked structure; it establishes the order of magnitude of t at van der Waals contact.

Reorganization energy

The inner-sphere reorganization energy of the back transfer, $\text{Fl}^{\bullet-} + \text{amine}^{\bullet+} \rightarrow \text{Fl} + \text{amine}$, was obtained from the four-point scheme¹³ with B3LYP energies at GFN2-xTB geometries:¹⁴ $\lambda_i = 0.63$ eV with methylamine, 0.48 eV with benzylamine and 0.92 eV with alanine, of which 0.20 eV comes from the flavin. Adding an outer-sphere contribution of a few tenths of an electronvolt for a buried active site gives $\lambda \approx 0.7\text{--}1.4$ eV, the range used below. It brackets the reorganization energies measured for flavin–tryptophan electron transfer in photolyase, 0.95–0.98 eV for charge recombination of the $\text{FAD}^{\bullet-}/\text{W382}^{\bullet+}$ pair.¹⁵

A measured anchor

The same photolyase study measured the recombination of flavin radical pairs at contact directly: $\text{FAD}^{\bullet-}$ with $\text{W382}^{\bullet+}$ recombines in 70 ps and with $\text{W384}^{\bullet+}$ in 120 ps, in the Marcus inverted region with a driving force of about 1.8 eV.¹⁵ Read through the Marcus expression below, the 70 and 120 ps lifetimes correspond to $t \approx 22$ and 4 meV, below our computed range, as expected for an edge-on rather than face-on donor–acceptor contact. We use it as the long-lifetime, weak-coupling end of the calibration.

One coupling sets both J and τ

The radical pair lies an energy Δ above the closed-shell ground state ($\text{Fl} + \text{amine}$) and is connected to it by the one-electron coupling t . The singlet pair mixes with the ground state through a matrix element $\sqrt{2}t$; the triplet cannot. To second order the singlet is raised by $2t^2/\Delta$, so

$$|J| = \frac{2t^2}{\Delta}. \quad (3)$$

The same coupling drives back electron transfer, $\Delta G = -\Delta$, at the non-adiabatic rate

$$k_{\text{BET}} = \frac{2\pi}{\hbar} t^2 \text{FC}, \quad \text{FC} = \frac{1}{\sqrt{4\pi\lambda k_B T}} \exp\left[-\frac{(\lambda - \Delta)^2}{4\lambda k_B T}\right]. \quad (4)$$

Their ratio contains no t :

$$\frac{|J|}{\hbar} \tau_{\text{BET}} = \frac{1}{\pi \Delta \text{FC}}. \quad (5)$$

The number of exchange precessions the pair completes before it recombines is fixed by Δ and λ alone. It is smallest near the activationless point, where it is 0.18 for $\Delta = \lambda = 1$ eV, and it grows steeply into the inverted region: 57 for the photolyase parameters. The coupling—the least certain input—cancels. A radical pair with a long lifetime is therefore necessarily one with a small t , and so a small J ; a pair with a strong coupling is necessarily short-lived. For large t the golden-rule rate exceeds nuclear frequencies, and we cap it by interpolating to the adiabatic limit, $k^{-1} = k_{\text{BET}}^{-1} + k_{\text{ad}}^{-1}$ with $k_{\text{ad}} = \nu_n \exp[-(\lambda - \Delta)^2/4\lambda k_B T]$ and $\nu_n = 10^{13} \text{ s}^{-1}$. The cap lengthens lifetimes and so errs towards magnetosensitivity.

Catalytic competence bounds the lifetime

A radical pair that is an intermediate of catalysis must form at least as fast as the enzyme turns over. In the dark it can only form thermally, by electron transfer up the same energy gap Δ , and by detailed balance its formation rate is the Marcus rate with $\Delta G = +\Delta$. In monoamine oxidase, steady-state turnover and anaerobic flavin reduction proceed at similar rates,¹⁶ so a catalytic intermediate must form at least at the flavin-reduction rate. Requiring the thermal formation rate to reach any value between 1 and 100 s^{-1} gives an upper bound on Δ . The bound depends on the required rate only logarithmically, shifting by $k_B T \ln 100 \approx 0.12 \text{ eV}$ across the two decades, so its conclusion does not hinge on the precise turnover number. Over $t = 22\text{--}290 \text{ meV}$, $\lambda = 0.7\text{--}1.4 \text{ eV}$ and $k_{\text{cat}} = 1\text{--}100 \text{ s}^{-1}$ the bound lies between 0.53 and 0.93 eV (Table 1). A pair this close to the ground state recombines in the Marcus

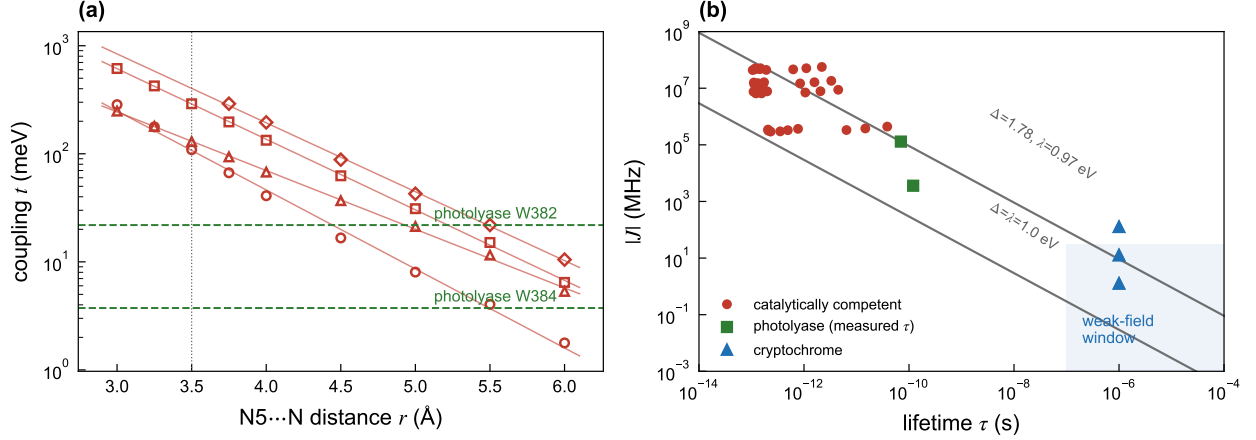


Figure 1: (a) Electronic coupling t between the flavin LUMO and the amine HOMO (B3LYP/def2-SVP, fragment projection) as a function of the N5...N distance, for four orientations of the amine; lines are exponential fits. Dashed lines: couplings implied by the measured recombination times of FAD $\bullet^-$ with W382 $\bullet^+$ and W384 $\bullet^+$ in photolyase.¹⁵ (b) Exchange coupling against lifetime. Grey lines: equation 5 for the photolyase parameters and for the activationless case. Red: radical pairs that form thermally at 1–100 s $^{-1}$, evaluated at the bound $\Delta_{\max}$ for all couplings and reorganization energies of Table 1. Green: photolyase. Blue: cryptochrome at 1.90 nm over the literature range of the exchange coupling.⁹ Shaded: the region in which a 50 μ T field can act, $\tau \gtrsim 0.1$ μ s and $|J|$ below the hyperfine scale.

normal region: at the bound its lifetime is between 1.1×10^{-13} and 3.9×10^{-11} s, and equation 3 gives $|J|$ between 2.9×10^5 and 5.6×10^7 MHz.

The argument is conditional in the useful direction. If the radical pair is on the catalytic pathway, it lives for tens of picoseconds at most and is exchange-locked. If it is long lived, it cannot form at the catalytic rate and does not take part in turnover. The long lifetimes that weak-field sensitivity requires, $\tau \gtrsim 100$ ns, are available to a thermally formed contact pair only in the second case. Mechanistic studies of monoamine oxidase in any event favour a polar mechanism over one-electron transfer,^{16–18} which places the enzyme in the second case already; the bound above shows that the conclusion does not rest on that mechanistic assignment.

Magnetic field effects

The contact dipolar coupling

At the active-site contact distance the point-dipole approximation gives $D = -1817$ MHz for MAO (3.5 Å) and -1217 MHz for DAO (4.0 Å). At contact the approximation overstates the coupling, because the flavin spin is spread over the isoalloxazine ring. Summing the dipolar tensor over the atomic spin populations of the two radicals, as Efimova and Hore did for cryptochrome,⁹ gives $D = -914$ and -674 MHz, about half the point-dipole values and still several times the largest hyperfine coupling in either pair. On its own this coupling limits the geomagnetic response. With J set to zero, lifetimes of 1–100 μ s and the partner hyperfine coupling anywhere in 5–363 MHz, the coherent MFE does not exceed 4.3×10^{-4} % for MAO and 4.1×10^{-4} % for DAO at the field orientations examined (0, 45 and 90° to the interrational axis). This is the most favourable case the dipolar coupling allows, and it ignores the competence bound, which by itself removes any effect. The exchange coupling and short lifetime required by catalytic competence reduce it further, to below the precision of the calculation (Supporting Information, Figure S1).

Cryptochrome in the same model

The contrast is not a property of the flavin. The cryptochrome radical pair, FAD $\bullet^-$ with the terminal tryptophan W324 at 1.90 nm, has $D = -11.4$ MHz and an exchange coupling of comparable magnitude and uncertain sign.⁹ Evaluated with the same Hamiltonian over the full literature range of the exchange prefactor and both signs, with a 1 μ s lifetime and $T_2^e = 1$ μ s,¹⁹ its geomagnetic response is 0.21–1.74 % with the field perpendicular to the interrational axis and 0.18–1.42 % averaged over orientations. Treating ^{14}N as spin 1 changes these values by at most a factor of 2.0. The difference from the enzymes is not the cofactor but how the pair is made. Light supplies the driving force, so the pair forms in under a picosecond while lying far from the ground state; charge separation along the tryptophan

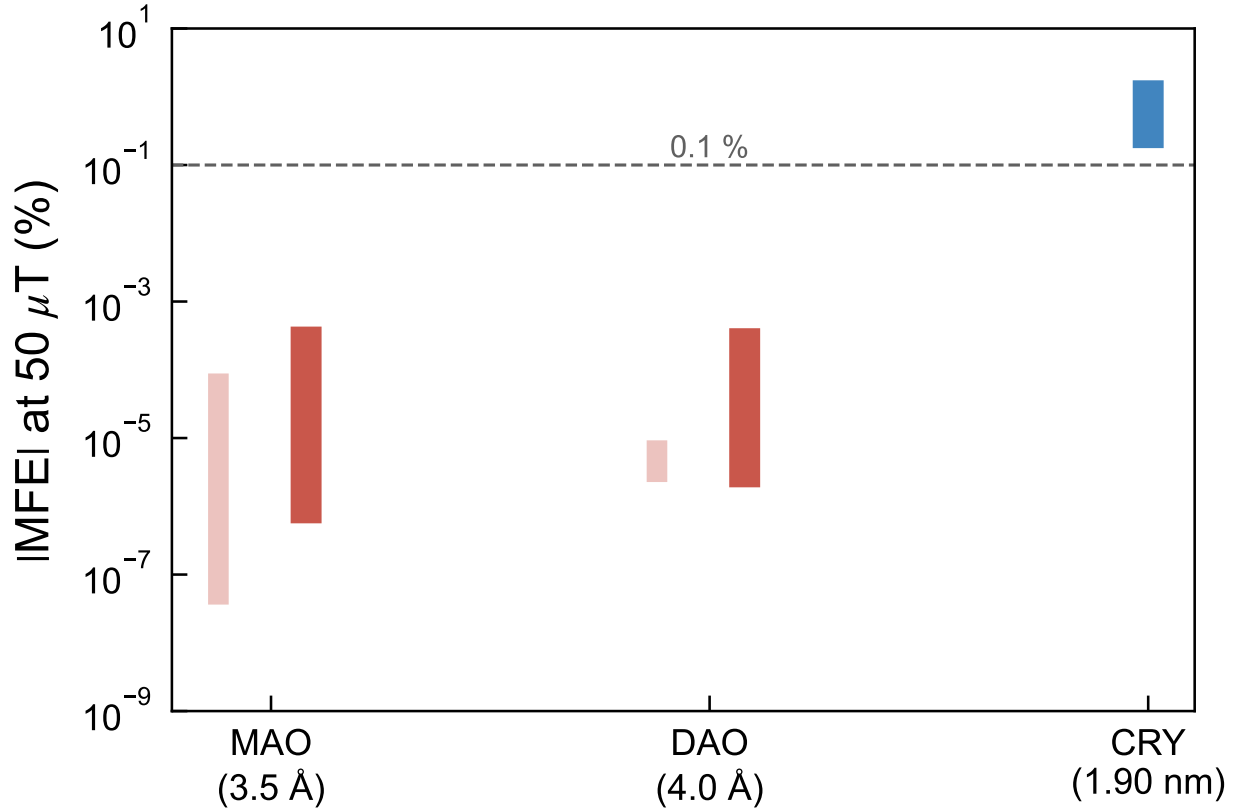


Figure 2: Geomagnetic-field effect on the singlet yield. MAO and DAO: range of the coherent MFE with $J = 0$, lifetimes of 1–100 μ s, field orientations 0–90° and partner hyperfine couplings of 5–363 MHz (MAO) or 5–100 MHz (DAO)—the most favourable conditions the contact dipolar coupling allows. Dark bars: dipolar coupling summed over the DFT spin distributions; light bars: point dipole. Cryptochrome: range over the literature exchange prefactor and both signs, spin- $\frac{1}{2}$ and spin-1 nitrogen models, perpendicular field and orientation average. Dashed line: 0.1 % for reference.

chain then moves the hole away, reducing t and with it both J and the recombination rate. A dark enzyme cannot do this, because the energy it would need to form a well-separated, long-lived pair is the energy that detailed balance forbids it to borrow at the catalytic rate.

Discussion

Three limitations should be stated. The coupling t was computed for a model placement of the substrate rather than a docked or simulated enzyme–substrate complex, and the inner-sphere λ from gas-phase energies at semiempirical geometries. Both enter only through ranges, and equation 5 removes t from the relation between J and τ ; the competence bound is insensitive to either over the ranges examined. Second, the non-adiabatic/adiabatic interpolation is approximate, but it errs towards longer lifetimes, and the dipolar coupling is taken from atomic spin populations rather than the full spin density; the bound it gives changes by an order of magnitude between the point-dipole and spin-population values, and neither approaches a measurable effect. Third, we have not treated three-radical mechanisms,²⁰ in which a third spin restores sensitivity to an otherwise inert pair. These require the pair to survive long enough to encounter the third radical, which the competence bound excludes for a catalytic intermediate.

The analysis applies to any radical pair that forms thermally and recombines to a closed-shell state, not only to monoamine oxidase. It suggests a simple classification: flavin radical pairs that are formed by light can be magnetosensitive, because the photon pays for the separation that decouples them, whereas pairs formed in the dark as catalytic intermediates cannot.

Conclusions

For a flavin radical pair that recombines to its ground state, the exchange coupling and the lifetime are set by one electronic coupling, and their product depends only on the driv-

ing force and the reorganization energy. With these quantities calibrated for the flavin-amine pair proposed for monoamine oxidase and anchored to measured flavin radical-pair recombination, detailed balance confines any catalytically competent pair to lifetimes below 3.9×10^{-11} s and exchange couplings above 2.9×10^5 MHz, and the contact dipolar coupling alone would limit its geomagnetic response to 4.3×10^{-4} % even without that bound. The neurotransmitter-metabolizing flavoenzymes cannot sense the geomagnetic field through the radical-pair mechanism; cryptochrome can, because light, not the enzyme, pays for the separation that its magnetosensitivity requires.

Table 1: Upper bound $\Delta_{\max}$ on the driving force for a radical pair that forms thermally at the rate $k_{\text{cat}} = 10 \text{ s}^{-1}$, with the lifetime and exchange coupling at the bound. The full grid ($k_{\text{cat}} = 1\text{--}100 \text{ s}^{-1}$) is in the Supporting Information.

t (meV)	λ (eV)	k_{cat} (s^{-1})	$\Delta_{\max}$ (eV)	τ (s)	$ J $ (MHz)
22	0.7	10	0.74	2.1×10^{-13}	3.2×10^5
22	1.0	10	0.71	4.9×10^{-13}	3.3×10^5
22	1.4	10	0.62	1.5×10^{-11}	3.8×10^5
109	0.7	10	0.82	1.3×10^{-13}	7.0×10^6
109	1.0	10	0.81	1.5×10^{-13}	7.1×10^6
109	1.4	10	0.73	2.1×10^{-12}	7.8×10^6
160	0.7	10	0.84	1.3×10^{-13}	1.5×10^7
160	1.0	10	0.83	1.3×10^{-13}	1.5×10^7
160	1.4	10	0.76	1.6×10^{-12}	1.6×10^7
290	0.7	10	0.86	1.4×10^{-13}	4.7×10^7
290	1.0	10	0.87	1.2×10^{-13}	4.7×10^7
290	1.4	10	0.80	1.1×10^{-12}	5.1×10^7

Supporting Information Available

Coupling scan geometries and fits; four-point energies; hyperfine couplings of all nuclei; the magnetic field effect along the coupling-locked (J , τ) locus; partner hyperfine and field-orientation dependence; spin-1 nitrogen model; software and environment.

Data availability

All code and numerical results are available in the open-source `qbsscreen` package at https://github.com/deeptell-inc/brain_protein_screening; every number in the text is generated from the stored results by `qbsscreen/make_numbers_tex.py`.

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TOC Graphic

